

Free Enrichment in Free Algebras

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We seek to categorically explain a construction that one encounters in program verification. One usually starts by fixing a (denotational) semantics for programs in terms of a traced distributive monoidal category $\mathbb{C}$: the semantics of a program is a morphism in $\mathbb{C}$. Usually, the category $\mathbb{C}$ is some Kleisli category of a $\mathbf{Set}$ -monad, $\mathbf{T}$, so its objects are just sets. Then, one wants to reason about properties of the outputs of programs given that the inputs satisfy some other properties. A property, P , on a set, A , is often defined as a subset of $\mathbf{T}$ -effects on A , $P \subseteq \mathbf{T}(A)$, rather than as a fuzzy predicate in the Kleisli category $\mathbf{kl}(\mathbf{T})$ [den99,dd02,MMS96,BBKW24,Kam18]. This note seeks to answer the following question.

Why are properties defined as subsets of effects?

The answer that we propose relies on considering the free enrichment of $\mathbb{C}$ on the free algebras of the powerset monad. We recall some facts about changing the base of enrichment; these can be found in Cruttwell's thesis [Cru08].

Change of base of enrichment. A lax-monoidal functor, $F: \mathbb{V} \rightarrow \mathbb{W}$, between two monoidal categories, $\mathbb{V}$ and $\mathbb{W}$, determines a 2-functor, $F_*: \mathbb{V} - \mathbf{Cat} \rightarrow \mathbb{W} - \mathbf{Cat}$, from the 2-category of $\mathbb{V}$ -enriched categories to that of $\mathbb{W}$ -enriched categories. This construction can be found in Section 4.2 of Cruttwell's thesis [Cru08] and is originally by Eilenberg and Kelly [EK66].

Intuitively, this 2-functor changes the base of enrichment; we describe it explicitly on objects. For a $\mathbb{V}$ -category, $\mathbb{C}$, we define a $\mathbb{W}$ -category, $F_*(\mathbb{C})$. The set of object of $F_*(\mathbb{C})$ is the same as the set of objects of $\mathbb{C}$; for two objects, A and B , the hom-object is defined by postcomposing with the functor F , $F_*(\mathbb{C})(A, B) := F(\mathbb{C}(A, B))$; the composition and identities are defined by monoidally applying F , $\circ_{A,B,C}^{F_*\mathbb{C}} := m_{\mathbb{C}(A,B), \mathbb{C}(B,C)} \circ F(\circ_{A,B,C}^{\mathbb{C}})$ and $\mathbb{1}_A^{F_*\mathbb{C}} := u \circ F(\mathbb{1}_A^{\mathbb{C}})$, where $m_{X,Y}: FX \otimes FY \rightarrow F(X \otimes Y)$ and $u: I \rightarrow F(I)$ are the monoidal structure of the functor F .

If the lax-monoidal functor F is strong monoidal and has a right adjoint as a functor, i.e. the adjunction is monoidal, then F_* also has a right 2-adjoint [Cru08, Section 4.4]. If the $\mathbb{V}$ -category $\mathbb{C}$ has a monoidal structure relative to $\mathbb{V}$, then $F_*(\mathbb{C})$ has a monoidal structure relative to $\mathbb{W}$ [Cru08, Chapter 5].

Free enrichment in free algebras. Consider a lax-monoidal monad, $\bar{F}: \mathbb{V} \rightarrow \mathbb{V}$, and the adjunction with its category of free algebras, $F: \mathbb{V} \rightleftarrows \mathbf{FAlg}_{\bar{F}}: \mathbf{U}$. This adjunction is always monoidal because the functor F is strong monoidal [Sea13, Remark 2.7.3]. By the constructions above, we obtain a 2-adjunction $F_*: \mathbb{V} - \mathbf{Cat} \rightleftarrows \mathbf{FAlg}_{\bar{F}} - \mathbf{Cat}: \mathbf{U}_*$.

Note that, if we considered the adjunction with the Kleisli category, we would obtain a $\mathbf{kl}(\bar{F})$ -enriched category that is isomorphic to the original one. This is because the functor $K: \mathbb{V} \rightarrow \mathbf{kl}(\bar{F})$ is identity on objects while the functor $F: \mathbb{V} \rightarrow \mathbf{FAlg}_{\bar{F}}$ is not. However, the 2-categories $\mathbf{FAlg}_{\bar{F}} - \mathbf{Cat}$ and $\mathbf{kl}(\bar{F}) - \mathbf{Cat}$ should be isomorphic, so considering morphisms $F_*\mathbb{C} \rightarrow F_*\mathbb{D}$ in $\mathbf{FAlg}_{\bar{F}} - \mathbf{Cat}$ should be the same as considering morphisms $\mathbb{C} \rightarrow \mathbb{D}$ in $\mathbf{kl}(\bar{F}) - \mathbf{Cat}$ (and not the same as morphisms $\mathbb{C} \rightarrow \mathbb{D}$ in $\mathbb{V} - \mathbf{Cat}$).

Free enrichment in free powerset-algebras. We are interested in the case of the powerset monad, $\mathbb{V} = \mathbf{Set}$ and $F = \wp$; the free algebras of the powerset monad are free semilattices. For a ($\mathbf{Set}$ -enriched) category $\mathbb{C}$, we describe $\wp_*\mathbb{C}$ explicitly. The objects of $\wp_*\mathbb{C}$ are the objects of $\mathbb{C}$. The sets underlying the hom-objects are $(\wp_*\mathbb{C})(A, B) := \wp(\mathbb{C}(A, B)) = \{F : F \subseteq \mathbb{C}(A, B)\}$; their semilattice structure is given by subset inclusions and unions. For two morphisms, $F \in (\wp_*\mathbb{C})(A, B)$ and $G \in (\wp_*\mathbb{C})(B, C)$, their composition is $F \circ G := \{f \circ g : f \in F \wedge g \in G\}$. Identities are singletons, $\mathbb{1}_A^* := \{\mathbb{1}_A\}$.

If the category $\mathbb{C}$ is monoidal, then so is $\wp_*\mathbb{C}$. On objects, the monoidal product of $\wp_*\mathbb{C}$ is defined as that of $\mathbb{C}$; on morphisms, it is defined pointwise, $F \otimes G := \{f \otimes g : f \in F \wedge g \in G\}$. It is easy to check that bimonoidal structures and traced structures are also preserved by $\wp_*$ because the inclusion $\mathbb{C} \rightarrow \wp_*\mathbb{C}$ given by taking singletons preserves naturality. Note that naturality of the monoid structure of coproducts is not preserved: the morphisms that do commute with the monoid structure are precisely the singletons, i.e. the ones that come from $\mathbb{C}$.

We conclude by observing that properties on an object A of $\mathbb{C}$ should be defined as states $1 \rightarrow A$ in the $\mathbf{FSLat}$ -category $\wp_*\mathbb{C}$: when $\mathbb{C}$ is the Kleisli category of a $\mathbf{Set}$ -monad T , states $1 \rightarrow A$ in $\wp_*\mathbf{kl}(T)$ are subsets $P \subseteq T(A)$.

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